



Ecological wars and their impact on international relations in the post-Cold War world.1991-2022

*rima djelloul **

*Mohamed Lamine Debaghine,Setif-02-
ri.djelloul@univ-setif2.dz*

k̄arima abassi

*Mohamed Lamine Debaghine,Setif-02-
k.abassi@univ-setif2.dz*

Abstract

The study of ecological issues in the post-Cold War world aims to present the most prominent discussions and points of view on the impact of environmental issues on international relations, given that the environmental variable is of great importance in terms of its relationship to various international issues related to global security, whether at the level of the state or at the level of the individual.

Therefore, research in the field of ecological warfare in the period extending from 1991 to 2022 is no longer merely an incidental result of international conflicts and rivalries, but rather has become a means and a strategy in itself, which requires a reconsideration of the course of the international security environment and the concepts of power, national and environmental security, sovereignty, and international borders, which are considered important issues in the twenty-first century.

Article info

*Received
Septembre 03.2025*

*Accepted
December 30.2025*

Keyword:

- ✓ *ecological wars,*
- ✓ *ecological issues*
- ✓ *environmental variable,*

* Corresponding author

1. **Introduction**

Ecological wars were not a focus of attention in the field of international relations during the Cold War period, as all issues focused on armed conflicts and disputes, the arms race, the mobilisation of military and nuclear power and the exploitation of spheres of influence and geographical space - and if there were voices calling for consideration of the environment and ecological wars, they were very limited and dominated by military perceptions.

With the end of the Cold War, the disintegration of the Soviet Union and the emergence of a new international pattern led by the United States of America, which was accompanied by technological developments that brought with them the beginnings of nuclear reactors and their effects on countries and neighbouring countries, the world realised that there are climate changes such as desertification, pollution and other climate fluctuations that caused the disruption of the ecological balance.

In this regard, environmental issues have become at an accelerated pace in parallel with the technological development of industrial companies that threaten the safety and security of individuals on the surface of the globe, which resulted in many crises that do not recognise international borders, such as health, food, famine, floods and others, which led to the concerted international efforts to eliminate this by establishing various international agreements and treaties to limit this.

With the emergence of the Corona pandemic and its rapid spread throughout the world, factories stopped working, and transport movements of all kinds stopped to take precautionary measures to limit the spread of the virus, and this reflected positively on the

environment in all its dimensions, and therefore the following problematic came:

To what extent have ecological wars become a factor in the field of international relations in the post-Cold War world?

To answer the problem raised, we will address the following topics:

First chapter : Theoretical and Conceptual Framework for Ecological Warfare

Second chapter : The impact of ecological wars on international relations

Third chapter : Ecological issues in the context of the coronavirus pandemic

First chapter :

Theoretical and Conceptual Framework for Ecological Warfare

Notably, at the beginning of the twenty-first century, environmental issues received firm attention in areas of theoretical and empirical work, whether by political leaders or government officials, especially after the United Nations conference in Rio de Janeiro, which led to international and regional movements at the level of international institutions to review the paths of security action as an environmental challenge requiring the development of an international discipline to study the global environmental variable. Our study therefore poses a paramount importance on the extent to which environmental security can be achieved in light of the rising threat of ecological wars, based mainly on the recognition of the nature of the interrelated environmental and security issues, summarised as follows.

In addition, ecological wars have contributed to reshaping the priorities of international and global security, as the focus has shifted from traditional military threats to international

security to environmental dimensions and their multilateral and multi-sectoral repercussions, which have given a new dimension to the concept of international security. This is evident in the innovative strategies adopted by countries in managing environmental conditions, including environmental destruction, pollution, tensions, and human displacement resulting from environmental destruction. This will be addressed through the following:

The first Subchapter: The concept of ecological warfare

Ecological wars are defined as imperialist wars that come in the form of a response to the interests and strategies of the great powers in defeating the enemy by employing a wide range of advanced technologies that can trigger floods, hurricanes, earthquakes and droughts, in order to defeat the enemy.

Ecological wars are considered one of the important issues in international relations, which express the development of forms of international conflicts and disputes that are directly linked to ecological factors, as scientists and philosophers consider it to be the war in which the environment is used as a primary means to launch international conflicts and disputes, and adopting the principle of threatening the vital interests of states with the intention of weakening the opponent or achieving political and social goals, by taking as a reference the destruction of ecosystems, causing damage to the vegetation cover and the deliberate destruction of environmental horizons

Coinciding with the development of the concept of war after the Cold War, the environment became one of the important tools in managing international wars and escalating them towards achieving the interests of the major powers, and therefore the colonial countries resort to using the environment in

their wars, we all remember the war of napalm bombs in Vietnam, which destroyed large areas of arable land, enriched uranium in Iraq, the escalation of deliberate burning of agricultural crops and forests, and the pollution of water sources and the draining of dams in Syria, as well as the deliberate burning of agricultural crops and forests. The threat of destroying dams and flooding cities in Ukraine as environmental issues that have worsened in various regions of the world with different degrees of severity depending on the quality of pollutants, which affected the immunity of the environment and thus the loss of its lines of defence and the ineffectiveness of its shields within the theatre of operations between the earth's crust and the atmosphere, which witnessed human interventions in their occurrence considered as security implications on the change of global politics².

Based on the above, ecological wars are defined as a criminal human activity that violates the principles of environmental justice aimed at achieving ecocide through the use of biotechnology and biological weapons to activate environmental disasters and deliberately spread diseases with the ability to change body functions without the individual's knowledge of this easier, more lethal, less expensive, and more difficult to detect.

It is noted that advances in biotechnology will be used as a weapon to deliberately spread disease and environmental disasters in the future. The likelihood may be low, but the stakes are clearly very high. Governments and organisations are concerned because the potential consequences for humanity could be catastrophic and difficult or impossible to treat.

The study of ecological warfare and its interactions in international politics starts from the premise that the failure to prevent the use of biological weapons would severely undermine existing international treaties

banning biological weapons, including the 1925 Geneva Protocol prohibiting the use of asphyxiating, poisonous or other gases, and the Protocol on Bacteriological Methods of Warfare, which prohibits the use of biological weapons. It would also weaken the 1972 Biological Weapons Convention, which criminalises the development, production, stockpiling, possession, retention and transfer of biological weapons.³

Second Subchapter: Concepts related to ecological warfare

A-Ecocide:

is a criminal human activity that violates the principles of environmental justice, deliberately launching a blatant attack on nature and the environment with the intention of causing widespread and long-term damage that is clearly excessive in relation to the overall scale of expected and directly tangible military and non-military gains.⁴

Genocide is a practice that involves numerous environmental practices by humans or local authorities of a given country, with the aim of causing serious harm to environmental interests and resources, through the spread of pollution, the elimination of a specific plant species, the deliberate cutting of trees and destruction of vegetation, or the use of explosives and nuclear tests, all of which contravene the mechanisms of international humanitarian law and the promotion of environmental protection. From a legal perspective, the international security environment is witnessing a continuous consolidation of international, regional, and global efforts to directly and indirectly pressure companies and government institutions to protect vegetation, soil, and ecosystems from environmental damage and disruption, and to bring serious charges against perpetrators of environmental crimes.

B- Environmental cleansing:

It is a criminal operation aimed at eliminating forests and natural plants with the aim of distorting the environment and nature by various ways and means, including: poisoning trees, cutting down trees and rare natural plants, resulting in natural disasters that affect the peace and security of the individual.

It also refers to those governmental and non-governmental operations that aim to eliminate some plants in order to achieve political, military and economic interests and goals, including: cutting down trees and removing some plants from certain areas in order to build schools, hospitals and large factories to convert harmful waste and other illegal matters that harm the environment. This environmental cleansing differs from genocide in this way.

C- Ecological Weapons:

Ecological weapons are referred to as non-conventional weapons that include nuclear, radiological and chemical elements and a type of mass destruction that targets the components of the ecosystem of the environment with the intention of causing serious damage to the environment and nature.⁵

It is also considered an illegal technology and program that is deliberately used to destroy environmental facilities, ecosystems, and vital resources in order to achieve political and military goals and gains. This type of weapon depends on employing the environment itself in the field of conflict and international wars.

D-Compensation for ecological damage:

It is a monetary compensation based on the value of the damage caused to the environment or the environmental element with the intention of restoring the natural element to what it was before the damage occurred.⁶

It also refers to the legal and financial process that aims to repair environmental damage or restore ecosystems damaged as a result of illegal actions to ecosystems, especially those resulting from armed conflicts or unregulated industrial activities, in order to rehabilitate the environment.

E- Repairing Ecological Damage:

It is a process aimed at reconsidering environmental issues through the recommendation of governmental and non-governmental organisations by involving the environmental decision maker in this by setting legal mechanisms that spread environmental awareness. In order to impose a set of legal procedures and measures to improve the ability to recover from the total destruction that the environment was exposed to during pollution and severe environmental crises.⁷

F-Ecological Thinking: Interest in ecological thinking began in the early 1970s in conjunction with the emergence of environmental thinking, which is the link between the environment and security, to enhance understanding of the relationship between economic interdependencies, natural resources and trade to justify the redefinition of national security.⁸

G-The ecological print is a standardised, standardised measure of human demand on the Earth's ecosystems that reflects the state of resource demand relative to the Earth's ecological capacity to replenish its resources.⁹

Third subchapter: The green approach, the emergence of the need to explain ecological wars in international relations

Environmental issues and ecological wars are among the important issues in international relations and security studies, which made researchers focus on the acquisition of theoretical approaches concerned with

addressing environmental conditions and their impact on international interaction between various official and unofficial actors and institutions:

1- The emergence of interest in ecological wars in international relations:

Ecological and environmental issues in current international relations are of great academic and international interest due to the importance and seriousness of the emerging issues of global environmental change since the 1970s, inevitably reflecting an intellectual legacy beyond the collapse of the collective security aspirations of the League of Nations, and the beginnings and end of World War II as a response to projects to reform the international system through building cooperative institutions and developing international law by taking an approach to environmental policy issues under environmental legislation to reduce environmental hazards: Pollution, desertification, ozone layer hole, atmospheric inversion, which necessitated the need to unite international efforts to find solutions to them before they turn into acute international crises that pose a security threat to states.¹⁰

Political and economic currents also emerged in the form of social movements calling for the need for humanitarian intervention and the holding of international conferences on the need to preserve ecological and human security, including the Montreal Protocol of 1987 corresponding to the puncture of the ozone layer, which led to the emergence of ozone layer diplomacy, whose activities materialised within the framework of the so-called green policy, so environmental issues have reached great importance in linking the dimensions of human security and the environment.

After the Brundtland Conference and the Hague Declaration of 1989, international

cooperation activities emerged to meet the growing global demand for chemicals and their environmental impact on the sustainability of natural resources to enhance biodiversity to achieve environmental leadership.¹¹

After the Cold War, international relations experienced fundamental shifts in the content and processes of international relations, including the inclusion of environmental issues and environmental security as global issues that require the establishment of green movements and so-called green politics led by environmental cosmopolitanism and social movements, in addition to the academic debates among international relations theorists, especially in light of the fourth theoretical debate between rationalist, normative, and critical analyses, of which the green approach is one of the currents

2- The green approach and the explanatory power of the impact of ecological wars in international relations:

Green theory is considered an important paradigm shift in the field of international relations analysis, as it transformed the classical concern of the field of international relations from military security dominated by reliance on military power to cooperation between states in the highest issues within the framework of previous analyses towards concern for environmental issues such as climate change, and the breach of the ozone layer, where the ambition of green theory goes to understand and explain environmental issues as a process that affects the construction of international political power.

The study of these issues has increased especially after the growth of common problems between countries such as water scarcity, floods and others, as new issues that called for intensifying international co-operation efforts to eliminate them.

Second chapter

The impact of ecological wars on international relations

The end of the Cold War opened a door to change international conflicts and disputes, the conflict no longer revolves within the framework of the conflict between the great powers and the agency is no longer the decisive tool in the international arena, as it was assumed that the international conflict will take an environmental turn as a key pillar influencing national security policy, especially in the United States of America and therefore this axis will study the progress of environmental security as one of the concerns of national security policy and the possible consequences of linking national security policy to environmental issues.

First subchapter: Environmental security as a key determinant of national security policy

During the report issued by the Club of Rome in 1972, politicians and earth scientists noted the close link between security and the environment, especially with regard to the economic interrelationships between natural resources and trade, which reformulated national security by strengthening international perspectives after the 1987 World Commission on Environment and Development conference entitled Our Common Future.¹²

All political and national threats and environmental pressures are considered local, regional, and global influences that require non-military responses that called for the construction of an expanded security model that includes environmental threats and their implications. Along these lines, Kakkonen defined national security as "the tasks of the state that tries to ensure the security of its citizens against external threats." According to

Kakkonen, national security is achieved by using military force and direct means of confrontation to eliminate security threats with the ability to control local and international conditions.¹³

After the Cold War, the map of national security was expanded to include the environmental security model in its dimensions by relying on regional and international institutions to promote environmental well-being.

The second subchapter: The relationship between environmental security and national security

Testing the relationship between environmental security and national security requires examining the arguments of the critical approach that relies on the idea of broadening and deepening the concept of security, without going beyond

the international approach to security that calls for going beyond the military use of the international agenda, which refers us to discuss environmental and ecological issues as one of the most important determinants of national security, within the framework of the critical orientations accepted within security studies in the post-Cold War period, where the securitisation of environmental issues prevailed by formulating an approach that directs the debate towards focusing international relations on global rather than national security.¹⁴

Third chapter

Ecological issues in the context of the coronavirus pandemic

The spread of the Coronavirus has had a negative impact on many economic and financial sectors and fields, as well as other positive effects such as the Internet sector, e-commerce, e-services and others, and

considering that the environment is the most affected element of economic activities, the Coronavirus pandemic was a door for nature to breathe, as this axis explains.

First subchapter: Coronavirus pandemic: concept and management mechanisms

The Corona pandemic is known as a global crisis that plunged the world into an economic and political crisis that was considered the most severe and violent crisis that occurred in history, known as biological wars, this crisis led to the destruction of the infrastructure and superstructure of countries, which led to the development of all necessary measures in order to limit its spread.¹⁵

The essence of research into the Coronavirus pandemic is based on the development of frameworks that require a review of the measurement of the severity of the spread of the Coronavirus and its political and economic implications on the content of international politics, through the development of assessments that seek to understand the variation in the breadth and depth of the shifts from unipolar to multipolar system, and the systems responsible for managing current international affairs, and also included the study of the objectives of power and the role of the influence of prominent figures in global leadership, on knowing the most prominent political issues and actors as issues that need to be treated.¹⁶

The study of the spread of the Coronavirus aims to identify international mechanisms and strategies in combating the danger of the pandemic as an influential factor in international politics, its features have emerged in light of scientific and academic discussions in international relations, as after 2019, the Corona variable has become important in terms of its relationship with many international and environmental issues and the extent of its formation as an inevitable

danger that threatens the parameters of global security.

Second subchapter: Crisis management mechanisms in light of the Coronavirus outbreak

International crisis management is based on a set of international mechanisms, which are represented in the following:

A- Planning:

The importance of planning appears in the ability to extrapolate future events by relying on the techniques of future studies to predict the intention to deal effectively with the crisis, with the need to find all available alternatives while preventing and minimising future occurrences, Planning has become a crucial process to ensure the continuity of work for institutions, governments, and societies in a turbulent health and security environment. The epidemic has imposed on the health and economic levels the necessity of acquiring new strategies for management and solutions.

B- Technology: Smart technology works to provide information as quickly as possible in order to limit the spread of the crisis while providing all ways to contain it early, and this is reflected in the study of the Corona crisis model. Technology has played a major role in combating the coronavirus pandemic by advertising it and spreading media propaganda through social media and other means in order to contain the pandemic

C- Organisation: The ability to create a hierarchical climate capable of creating a consensus between different organisations and other international bodies in order to work intertwined in order to limit the crisis and the need to take the appropriate decision with the circumstances and challenges of the crisis.¹⁷

Third subchapter: Pandemic and its repercussions on the environment

Despite the negative effects of the Corona pandemic on the political and economic level, it has positively affected nature and the ecosystem. Due to the preventive measures taken by countries, pollution rates decreased due to the disruption of economic activities in various sectors, as scientific studies have found that the emission of carbon dioxide has decreased in many countries, including: European countries, the United States, and China, as in 2019, the global emission of 100 million tonnes of carbon dioxide as a result of burning fossil fuels. In 2019, the volume of global emission reached 100 million tonnes of carbon dioxide as a result of burning fossil fuels, with the beginning of 2020, gas emissions decreased to 83 tonnes, a decrease of 17%, while the emission level in some countries decreased by 26%, and the American News Agency recorded a decrease in air pollution in China as a result of the decline in economic activities in February compared to other states.¹⁸ Satellite data also showed a noticeable decrease in the level of air pollution, specifically in nitrogen dioxide gas during quarantine, as appeared slightly in northern Italy. In general, the European environment during the Corona pandemic witnessed a slight improvement in terms of decreasing the degree of air pollution

Conclusion:

The post-Cold War world has witnessed many changes, whether at the international or local level, including disasters and thorny security issues that called for the necessity of individual or collective action to eliminate and reduce them, by various means and strategies, according to the dynamics of their changing nature, and environmental and ecological issues are one of the most important new security challenges affecting international

peace and security, so states must take all precautionary measures to reduce them and their dangers.

According to the statistical study presented above, ecological issues are considered an influential factor in international relations and became of great importance during the post-Cold War period from 1991-2020, by placing environmental issues on the international agenda:

A-Results:

1- Ecological issues are considered one of the most important issues in the field of international relations, especially in the post-Cold War world, as they have become an indicator to measure the extent of cooperation and conflict between states to achieve their security.

2- Ecological issues in the post-Cold War world have become an important axis in theoretical debates, as raised by the fourth debate led by the green approach in international relations.

3- The student of international politics must understand the multiplicity and complexity of environmental aspects and the extent of their impact on changing the contents of power in international politics.

4- Environmental issues are global issues that call for concerted international efforts from governmental and non-governmental organisations that have the ability to control the paths of environmental fluctuations using the smartest programmes and strategies to reduce them.

5- Environmental security is considered a dimension of national security, as the destabilisation of the latter affects the security and stability of the national security of states.

B-Recommendations:

1-International research teams should be established for early detection, minimisation and treatment of environmental issues

2- The environmental development programme of the United Nations should be updated in order to preserve the security and safety of individuals and direct strict penalties for anyone who deliberately endangers the environment.

3- Encourage researchers in the field of environmental and developmental studies to innovate and discover the most prominent methods and strategies to reduce environmental risks.

4- Developing environmental advertisements and programmes to increase interest in ecological issues.

Bibliography:

First: References in Arabic:

Articles:

1-Mohamed, Rahmoun."Monetary compensation for ecological damage - a comparative study." Faculty of Law. **Taheri Mohamed Bechar University.**

2- Morsi ,Mohamed and Saltawi, Saltani, Adel Abdelhady Abdallah. "Compensation for environmental damage". **Institute of Environmental Studies - Sadat City University,** 2019.

3- Ahmed, Aboukasssem, Fakir Al-Bashir. "Environmental issues and their impact on international relations in the post-Cold War world from 1990-2020." **Alexandria University: Faculty of Economic Studies and Political Science.**

4- Ziani, Zidane. "The implications of environmental security on national security - a case study of Algeria". **Journal of the**

**Researcher in Political Studies. Issue: 02.
Volume: 09, 2022**

5- Mona, Znoudi. "Environmental Issues and the Need for Green Theorising in International Relations." **Al-Mufakkar Journal. Issue: 02. Volume: 14. June, 2019.**

6- Youssef, Lalgha. "Procedures for prosecuting perpetrators of environmental crimes before the International Criminal Court".

7- Laouina, Ben Zakoura. "The environment and the importance of clean technology in light of the Corona pandemic." **Journal of Advanced Economic Research. Issue: 01. Volume: 06. Year, 2021.**

B. Websites:

1-Imad Hatabeh, "Environmental Warfare at the Climate Conference", 25 November 2022, **<https://www.almayadeen.net>**

2-" Biotechnology, Weapons and Humanity - Frequently Asked Questions", 06-09-2002, **<https://www.icrc.org/ar/doc/resources/documents/misc/5n5bxc.htm>**

3- Noura, "International Politics and Corona Crisis Management: Experiences and expertise" **<https://eipss-eg.org>**

4-Mohammed Ahmed Al-Filabi, "Environmental Wars in the Present and Future", **<https://www.alaraby.co.uk>**

C. Foreign references :

1- "19 pandemic Humanity needs leadership and solidarity to defeat", **<https://www.lundp.org:covid>**

2-Greg Simons ." the coronavirus pandemic and Global transformation : Making or Breaking international orders" **[outline of global transformations, volum13, Number5, \(2020\)](#)**

3-**<https://www.who.int/health-topics/biological-weapons>**

4- "and advancing work to achieve environmental-leadership", **www.chemours.com/embracing**